How-To Library

Informatica

Calling a Web Service from a Web Service Consumer Transformation in Informatica Developer

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Abstract

In Informatica Developer, you can create a Web Service Consumer transformation that calls a web service from an Informatica Data Services mapping or an Informatica Data Quality mapping. This article describes the steps to create a Web Service Consumer transformation, configure the SOAP messages in the transformation, and connect the transformation to the web service.

Supported Versions

- Informatica Data Services 9.1.0
- Informatica Data Quality 9.1.0

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Overview

You can create a Web Service Consumer transformation that provides access to a web service from an Informatica Data Services mapping or an Informatica Data Quality mapping. The Web Service Consumer transformation can access an Informatica web service or a web service from an external provider.

Create a Web Service Consumer transformation from a Web Services Description Language (WSDL) data object. A WSDL data object is an object in the Model repository that defines a WSDL file. A WSDL file describes the web service operations and the request and the response messages.

The Web Service Consumer transformation can pass hierarchical groups of data to a web service in a request message. The transformation can receive multiple groups of data from a web service in a response message.

Scenario

Hypostores customer service representatives want to access customer data over a network. The customer service representatives need to view customer and order information by customer ID.

Use a Web Service Consumer transformation to receive a customer ID and pass the ID in a SOAP request to a web service. The web service returns the customer information and the customer orders in a SOAP response message. The Web Service Consumer transformation extracts one row of customer information and multiple rows of orders from the response message for each customer.

In Informatica Developer, create a mapping that includes a flat file source, the Web Service Consumer transformation, and the target customer and order tables.

Configuration Tasks

To implement this scenario, you must have a WSDL file that describes the web service you are going to connect to. Create the Web Service Consumer transformation from the WSDL.

1. Import a WSDL to create a physical data object.
2. Create the Web Service Consumer transformation.
3. Extract input ports to the SOAP request message.
4. Extract data from the SOAP response message to output ports.
5. Configure a connection to the web service.
6. Define the connection object name in the Web Service Consumer transformation.
7. Create the source and target physical data objects.
8. Create the Web Service Consumer mapping.
9. Run the mapping.

Step 1. Import a WSDL File

Before you create a Web Service Consumer transformation, the Model repository must contain a WSDL data object that describes the web service to run.

The WSDL describes the SOAP request message and the SOAP response message between the Web Service Consumer transformation and the web service provider. Import a WSDL file to create the data object in the Model repository. You can import the WSDL from the following location: https://communities.informatica.com/docs/DOC-5416.

1. In the Developer tool, click **File** > **New** > **Data Object.**
2. Select **WSDL Data Object** and click **Next.**

   The **New WSDL Data Object** dialog box appears.
3. Click **Browse** next to the **WSDL** option and enter the location of the WSDL. Click **OK.**

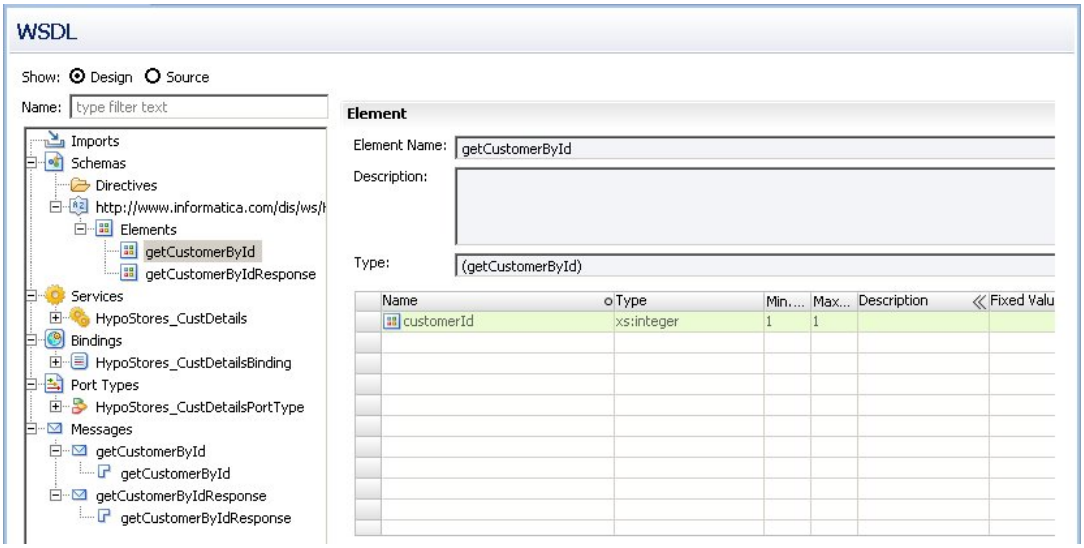
   You can enter a URI or browse for a file system location. The Developer tool validates the WSDL file. You can view the errors in the dialog box.
4. Enter a name for the WSDL.
5. Click **Browse** next to the **Location** option to select the project or folder location for WSDL data object.
6. Click **Next** to view the operations in the WSDL.

   For this scenario, the WSDL has one operation: getCustomerByID.
7. Click **Finish**.

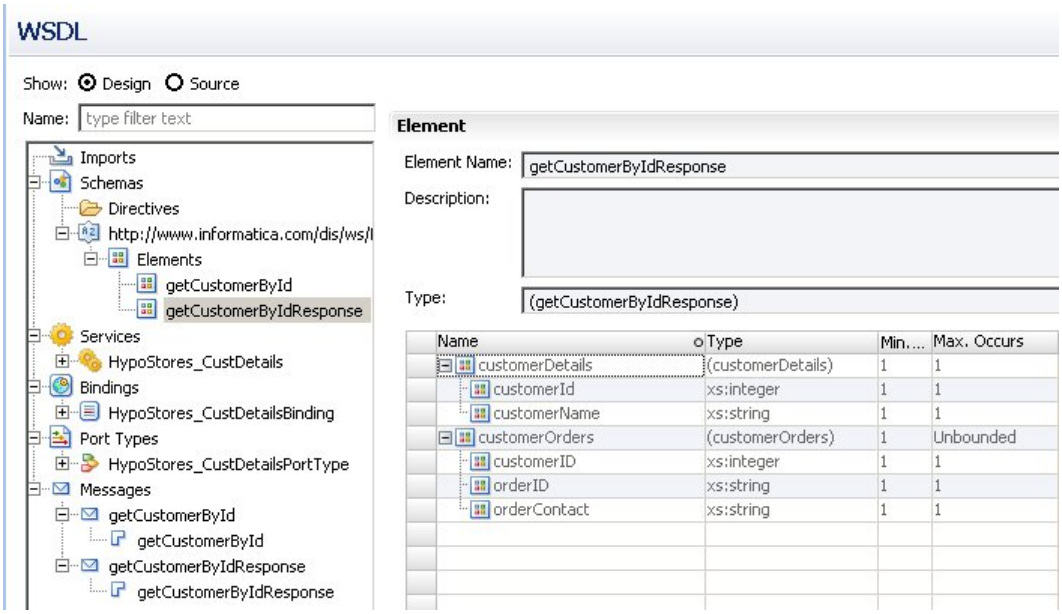
   The WSDL data object appears under **Physical Data Object** in the project or folder in the **Object Explorer** view. The **Overview** tab appears in the editor.

8. Click the **WSDL** tab and view the messages in the **Navigator** panel.

   The request message is getCustomerByID. The message contains a customer ID.



   The response message is getCustomerByIDResponse. The message contains a single-occurring CustomerDetails group and a multiple-occurring CustomerOrders group.



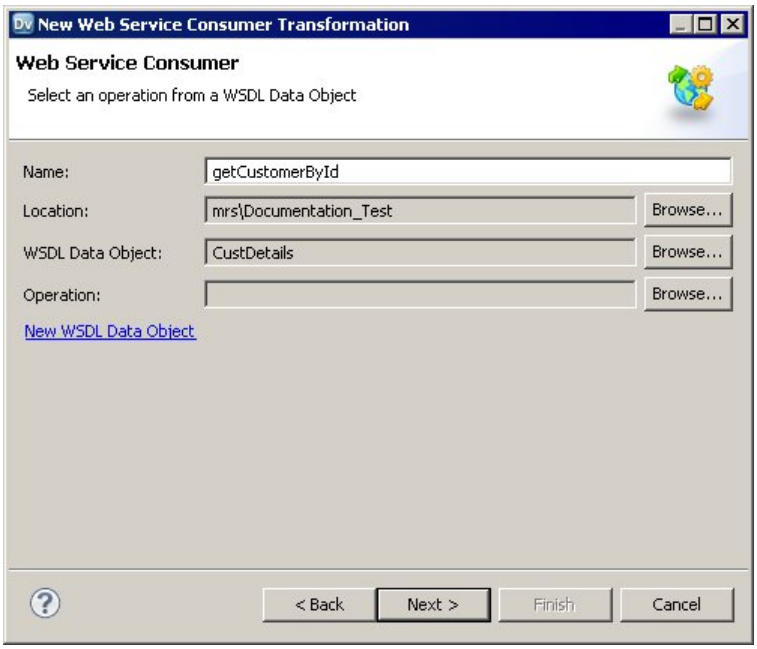
Step 2. Create the Web Service Consumer Transformation

Create a Web Service Consumer transformation from the WSDL data object.

Choose the web service operation that you want to run.

1. In the Developer tool, click **File** > **New** > **Transformation**.
2. Select Web Service Consumer and click **Next**.

   The **New Web Service Consumer** dialog box appears.

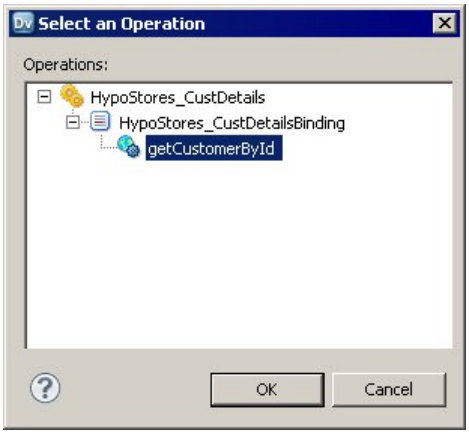


3. Browse and select a repository location for the transformation.

   The default is your current location.

4. Browse and select a WSDL data object to define the web service request and response messages.
5. Browse and select an operation from the WSDL.

   For this scenario, select the getCustomerByID operation.

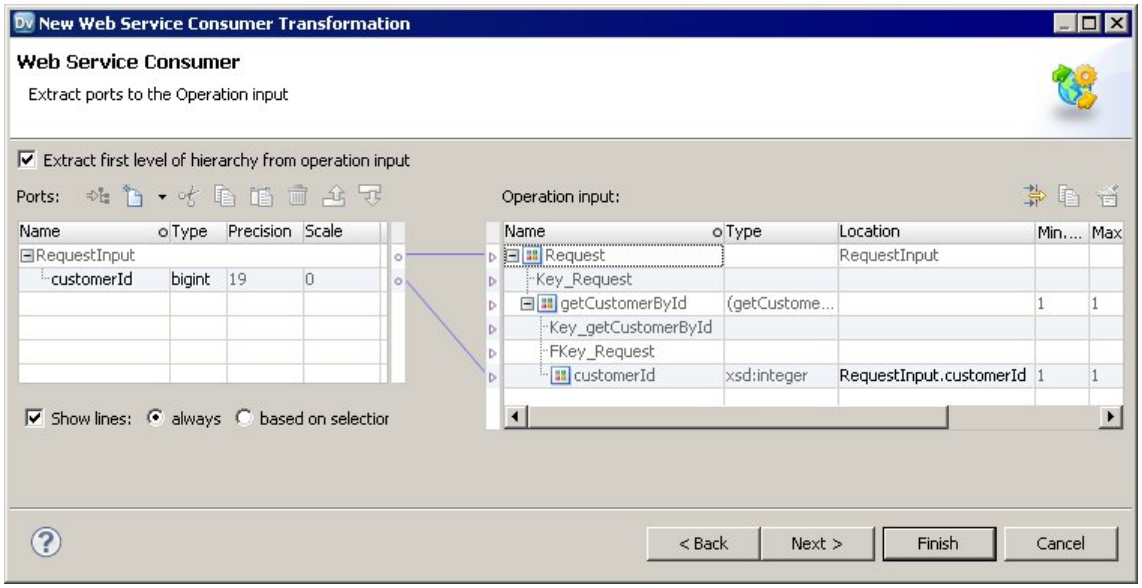


6. Click **OK**.

Step 3. Extract Ports to the Operation Input

Define the input ports and map where to extract the ports in the SOAP request message.

The following dialog box appears after you choose the getCustomerByID operation:



By default, the wizard creates the RequestInput port group. The wizard creates the customerID port and extracts the port to the customerID node in the Operation input. The wizard creates the input port when the **Extract the first level of the hierarchy from the operation input** is enabled.

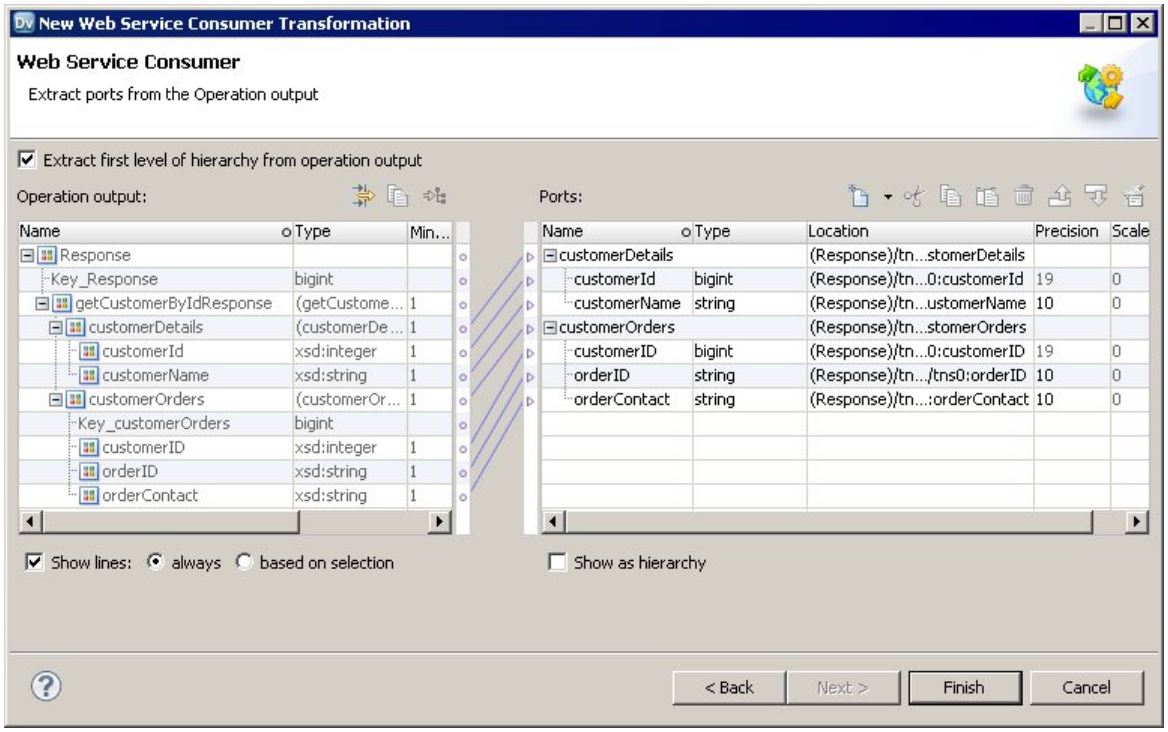
Note: You do not need to extract data for the keys in the Operation input hierarchy because the SOAP request contains one field.

Click **Next**.

Step 4. Extract Ports from the Operation Output

Create the output ports and extract the nodes from the SOAP response message to the ports.

The following dialog box appears after you extract ports to the Operation input:



The Operation output area contains the nodes in the SOAP response message. By default, the wizard creates the CustomerDetails group and the CustomerOrders group of output ports in the Ports area. The wizard extracts the nodes from the Operation output to the transformation output ports. The wizard creates the output port groups and extracts data from them when **Extract first level of hierarchy from operation output** is enabled.

Click **Finish** to create the Web Service Consumer transformation.

Step 5. Define a Connection to the Web Service

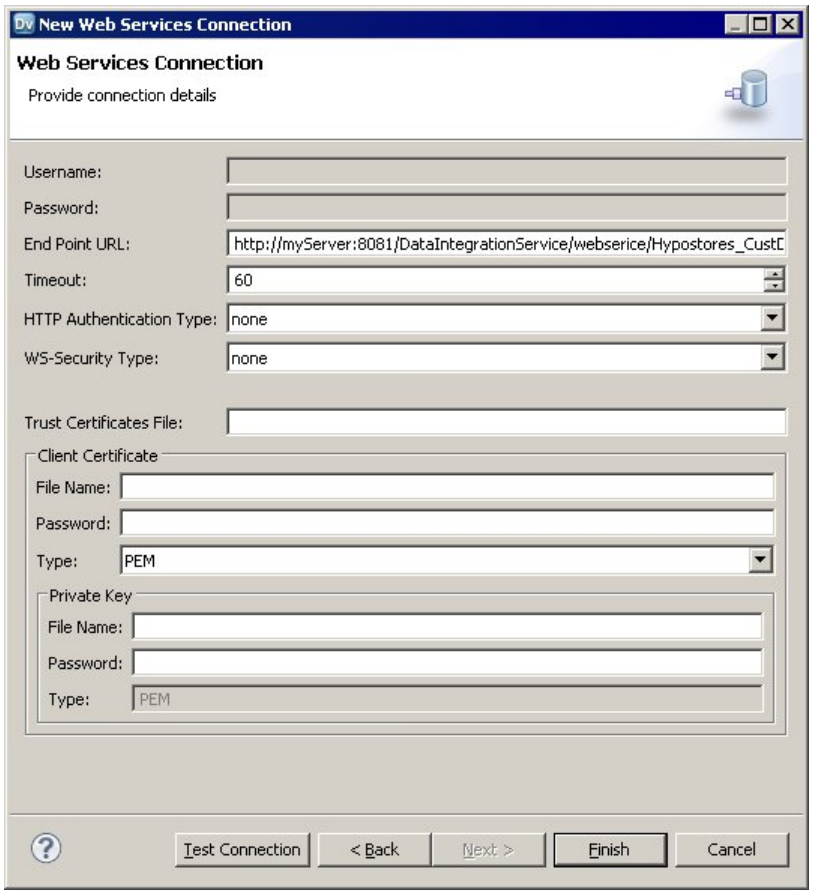
The Web Service Consumer transformation connects to a web service using the endpoint URL in the WSDL data object. The transformation can also connect to a web service using a web service connection. When you configure a web service connection, the connection overrides the URL in the WSDL.

Note: You must create a web service connection to configure WS-Security or to configure trust certificate files.

1. Click **Window** > **Preferences**.
2. Select **Informatica** > **Web Services** > **Connections**.
3. Select the domain and click **Add**.
4. Enter a connection name, description, and location.

   The connection type defaults to web services.
5. Click **Next**.
6. Enter the **End Point URL**.
7. Accept default values for the WS-Security Type and HTTP Authentication Type.

   For this example, the web service does not require user authentication.



Optionally, you can configure the following properties for a web service connection:

Property	Description
Username	User name to connect to the web service.
Password	Password for the user name.
End Point URL	URL for the web service that you want to access.
Timeout	Time in seconds that the Data Integration Service waits for response from the web service provider.

Property	Description
HTTP Authentication Type	Type of user authentication over HTTP. Select one of the following values: • None. No authentication. • Automatic. The Data Integration Service chooses the authentication type of the web service provider. • Basic. Requires you to provide a user name and password for the domain of the web service provider. The Data Integration Service sends the user name and the password to the web service provider for authentication. • Digest. Requires you to provide a user name and password for the domain of the web service provider. The Data Integration Service generates an encrypted message digest from the user name and password and sends it to the web service provider. The provider generates a temporary value for the user name and password and stores it in the Active Directory on the Domain Controller. It compares the value with the message digest. If they match, the web service provider authenticates you. • NTLM. Requires you to provide a domain name, server name, or default user name and password. The web service provider authenticates you based on the domain you are connected to. It gets the user name and password from the Windows Domain Controller and compares it with the user name and password that you provide. If they match, the web service provider authenticates you. NTLM authentication does not store encrypted passwords in the Active Directory on the Domain Controller.
WS Security Type	The WS-Security type that you want to use. Select PasswordText, PasswordDigest, or no WS-Security header.
Trust Certificates File	Trust certificates file name.
Client Certificate File Name	Client certificate file name.
Client Certificate Password	Client certificate password.
Client Certificate Type	The format of the client certificate file. Select one of the following values: • PEM. Files with the .pem extension. • DER. Files with the .cer or .der extension.
Private Key File Name	File name for the private key used when the web service provider authenticates the consumer or when the consumer and provider exchange certificates.
Private Key Password	Password for the private key file.
Private Key Type	The private key type is DER.

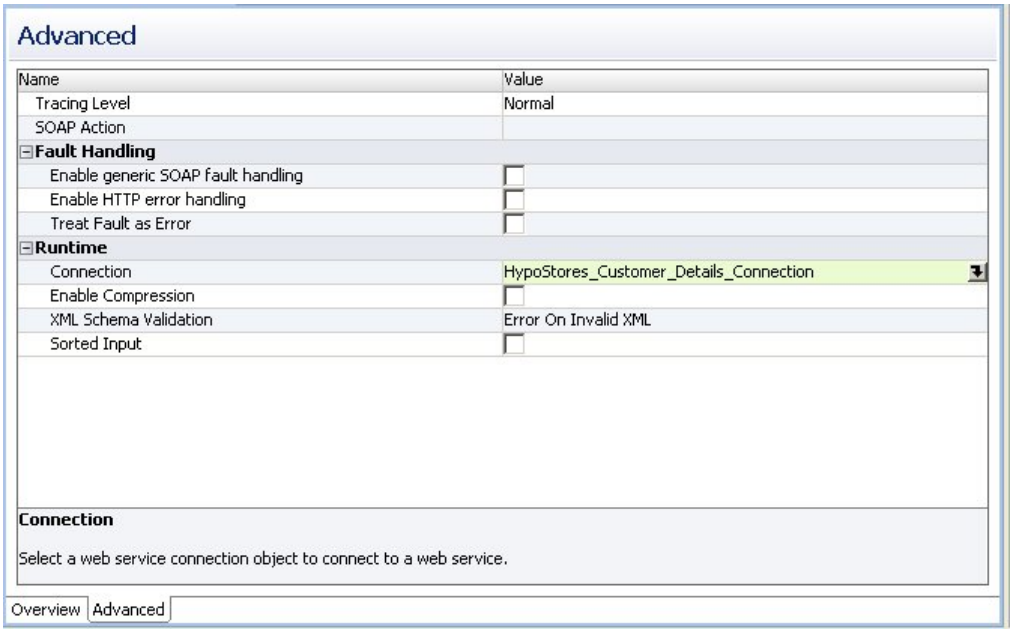
8. Click **Test Connection** to verify that you entered the connection properties correctly and that you can connect to the URI.
9. Click **Finish** to create the connection.

   You can choose this connection in the Web Service Consumer transformation.

Step 6. Configure the Connection in the Transformation Runtime Properties

Configure how the Web Service Consumer transformation connects to the web service. Select the web service connection that you defined in Step 5.

1. Click the Web Service Consumer transformation **Advanced** tab.
2. In the Runtime properties, browse for the web service connection object you created.



3. Save the Web Service Consumer transformation in the Model repository.

Step 7. Create the Physical Data Objects

Create a flat file data object source and relational data objects targets for the Web Service Consumer mapping.

1. Create a flat file data object.

   The flat file contains a list of customer numbers to retrieve orders for. Define the customer ID as an integer. Add the flat file source file name and directory to the Run-Time view on the Properties tab of the physical data object.
   The following list contains customer ID numbers:

   ```
   10110085
   10110086
   10110087
   10110088
   10110089
   ```

2. Define a Customer table and a Customer_Orders table to receive the data. Define the database connection for the tables.

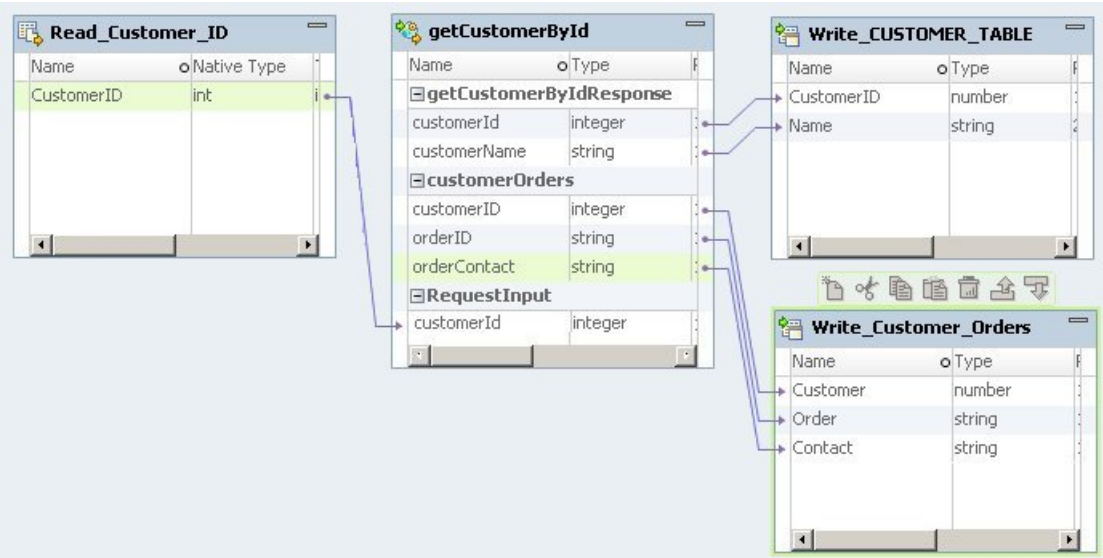
   The Customer table receives the CustomerID and Name.
   The Customer_Orders table receives the CustomerID, Order number, and Contact.

Step 8. Create the Mapping

Create a mapping with a flat file Input transformation, the Web Service Consumer transformation, and the Customer andthe Order relational tables that receive the data.

The flat file contains rows of customer numbers. For each input row, the Web Service Consumer transformation calls the getCustomerID web service. The web service returns customer information and multiple orders for the customer. The Web Service Consumer transformation passes the customer information and the orders to separate tables that are related by the customer ID.

The following figure shows the mapping:



1. To create a mapping, click **File** > **New** > **Mapping**.
2. Enter a name and location for the mapping.
3. Add the flat file physical data object to the mapping. Add the Web Service Consumer transformation to the mapping. Add the Customer and Order relational tables to the mapping.
4. Connect the input CustomerID to the customerID in the Web Service Consumer transformation.
5. Connect the output ports to the target tables. Connect the customerID and customerName ports from the getCustomerByIDResponse group to the Customer_Table. Connect the customerID, the orderID, and orderContact ports from the customerOrders group to the Customer_Orders table.
6. Click **Edit** > **Validate** to validate the mapping.

Step 9. Run the Mapping

Test the mapping to verify that the web service returns the expected response message. View the rows that the Customer or Order table receives from the Web Service Consumer transformation.

1. Click the Customer or Orders relational table in the mapping before you run the Data Viewer.
2. Click **Run** > **Data Viewer** to view the results.

When you click the Customer table, the Data Viewer shows the following rows:

CustomerID	Name
10110085	JOSEPHTHAL LYON & ROSS
10110086	NRCA
10110087	POND EQUITIES
10110088	SCHRODER & COMPANY
10110089	YUASA TRADING COMPANY (AMERICA)

When you click the Orders table, the Data Viewer shows the following rows:

Customer	Order	Contact
10110085	O-5079	PARKE PERSLEY OR RAYFORD LECROY
10110086	O-6658	ROLANDA SORTO
10110087	O-8195	KONSTANTIN PEDDICORD
10110088	O-9130	GIORGIA TWITCHELL
10110089	O-9352	STEFFI MCGLOWN

Author

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